LAYA: A Web-Based System for Automated Experimental Field Layout Generation, Visualization, and Planting Plan Report Generation

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I. Introduction

A. Background of the Study

Varietal trials are crucial to agricultural research since they enable the estimation and comparison of different crop varieties under controlled field situations. This is essential in varietal identification of high-yielding varieties that are resistant to diseases as well as those that are tolerant to adverse climatic conditions leading to food security as well as sustainability of crop production [16]. In varietal trials, experimental design accuracy plays a critical role in the credibility as well as reliability of results. In a bid to lower experimental error as well as account for environmental variability, scientists often make use of statistical field designs such as Randomized Complete Block Design (RCBD), which ensures that comparison of treatment is made within similar groups or blocks [3].

Fig. 1: Sample RCBD layout adapted from Khulna University (2020) [9].

In spite of its performance, the practical application of RCBD and other experimental designs is a challenge when dealing with several treatments, replications, and plot arrangements. The manual generation of layout as well as data administration is usually error-prone, inconsistent, as well as inefficient when it comes to varietal trials on a large scale [16] [7]. The manual approach not only wastes time but may also escalate mislabeling plots or losing varieties in the field, which can jeopardize accuracy in terms of analyzing as well as collecting data. As digital technologies as well as data-oriented approaches are becoming part of agricultural research informatics, automated systems that can simplify experimental layout generation as well as facilitate reproduction are needed with greater emphasis on overall data integrity [12].

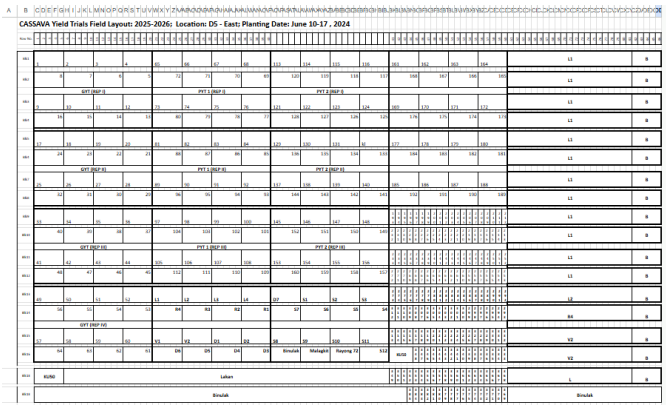

Fig. 2: Snapshot of manual RCBD layout from FIC-IPB (2025).

Additionally, traditional tabular or paper-based field layouts provide limited spatial understanding, which makes it hard for researchers to visualize the physical placement of plots across the field. The addition of field map visualization makes possible a more visual and spatially accurate presentation of experimental set-ups so that researchers can intuitively interpret field patterns without difficulty, verify randomization, as well as plan field management operations. With current web mapping technologies, these layouts can be interactively developed dynamically, visualized, and customized to represent accurately real-world field dimensions and interrelationships.

This research introduces Layout Assistant for Yield Assessments (LAYA), a web system developed for automated creation as well as visualization of experimental field designs. The system is intended to make it easier to create randomized experimental designs such as RCBD while presenting interactive visual maps that facilitate greater understanding and field experiment management in space. In bringing automation and visualization together on a single platform, LAYA aims to synchronize traditional experimental design approaches with contemporary digital tools to enable faster, more accurate and data-informed agricultural research.

B. Objectives

The primary objective of this study is to develop a web-based application that automates field layout generation and

field map visualization for varietal trials in the Institute of Plant Breeding. Specifically, it aims to:

1) Design and implement a modular web-based system architecture that supports multiple experimental designs, including the Randomized Complete Block Design (RCBD) and the Completely Randomized Design (CRD).
2) Automate the generation of accurate planting plans that include field dimensions, replication structures, and plot details, ensuring consistent and error-free documentation of varietal trials.
3) Develop and integrate interactive field map visualization features that allow users to display, manage, and validate field configurations dynamically through a user-friendly web interface.
4) Improve research efficiency and reproducibility by minimizing manual errors, streamlining layout generation, and supporting standardized documentation workflows.

C. Scope and Limitations

Scope

The present study is focused on the *LAYA* web-based application, specifically its development as an automatic system for the generation, randomization, and visualization of experimental field layouts commonly used in varietal trials of industrial crops. The system mainly supports the *Randomized Complete Block Design (RCBD)* and *Completely Randomized Design (CRD)* types and can be made adaptable for other designs in the future, such as the *Latin Square Design (LSD)* and *Split-Plot Design (SPD)*.

The proposed system will provide the following features:

1) Automatic randomization and layout generation according to the selected type of experimental design.
2) Visualization of field maps depicting the physical arrangement of the plots and replications.
3) Input of metadata and report generation, such as location of experiments, date of sowing, spacing, and other remarks.
4) Export of reports to PDF and spreadsheet, allowing printable planting plans and documentation for varietal trials.

The target users of the proposed system include researchers, scientists, field technicians, and data managers of the *Institute of Plant Breeding (IPB)* working under the *Feeds and Industrial Crops Division*, as well as other agricultural institutions conducting varietal trials.

Limitations

The system is limited to web-based deployment and requires a stable internet connection for proper functionality. The following features are not yet incorporated:

1) Statistical analysis features, such as analysis of variance (ANOVA) or means comparison tests.
2) Automated data collection through field sensors, mobile devices, or other electronic means.
3) Support for non-standard or mixed experimental designs that require customized layout configurations.
4) Offline functionality, as the system depends on browser-based processing and cloud-hosted components.
5) Advanced geo-spatial mapping capabilities, such as GPS integration, since the visualization is schematic rather than geospatial in nature.

It may also be noted that this system for layout generation, randomization, and visualization of experimental plots were evaluated primarily within the IPB context, and thus broader testing across different crops, institutions, or field scales may be required before wider adoption.

D. Significance of the Study

This study is significant as it addresses the automation and visualization needs of experimental field layouts used in varietal trials. It provides a valuable tool and sets of insights for researchers, field technicians, and plant breeders by integrating web-based technologies with spatial visualization frameworks such as Leaflet. The system is targeted at providing a more systematic, accessible, and interactive way to generate and manage experimental designs—a very important aspect of modern plant breeding and agricultural research.

In this system, plant breeders and researchers will have an efficient alternative to the traditional manual and desktop-based methods of layout generation, such as R's agricolae package. While these existing methods are statistically reliable, they require programming skills and often lack intuitive visualization. The current study introduces a web-based platform for the automatic generation of layouts for common designs such as *RCBD* and *CRD*. Visualization of plots in an interactive manner will allow users to more intuitively comprehend field arrangements and reduce errors at the setup and documentation stages of the varietal trial.

Compared to the existing approaches, LAYA offers quantifiable advantages. Automation of layout generation and report creation massively reduces the preparation time for field experiments–from hours or days to minutes. Interactive visualization further reduces setup and documentation errors, thus improving the reliability of trials. In addition, an integrated workflow in LAYA, including layout generation, spatial visualization, and report export, is rarely found in any of the existing tools that perform just one of these functions.

For a research institution such as the Institute of Plant Breeding, the system enhances standardization, reproducibility, and digital record keeping in experimental workflows. Automatically produced reports, including metadata on field dimensions, replication structures, and insertion of details in the documents, ease the documentation process by ensuring consistency between experiments. Such capabilities will let research programs become more productive, since an institution can now allocate its energy to the analysis of data and the making of decisions regarding breeding.

The system contributes, from the point of view of data managers and agricultural informatics specialists, to the growing development of digital agriculture by combining statistical

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design automation with web-based visualization. The system is highly modular, meaning it can be integrated with other tools like FieldBook and statistical software, allowing for a more comprehensive digital workflow: from the generation of experimental design to field data analysis. This study contributes to the digital transformation at a broader level, which involves agricultural research in the Philippines. By offering an open, scalable, and web-accessible platform for the design of experiments, researchers and institutions are encouraged to follow collaborative and data-driven practices. Ultimately, it aims to improve efficiency, transparency, and accuracy in varietal experimentation to ensure reliable research output, better selection of varieties, and sustainable development of agriculture.

II. Review of Related Literature

A. Experimental Designs in Plant Breeding

Field trials in plant breeding rely heavily on experimental designs that are structured to ensure fair evaluation of varieties under different field variability. Modern breeding programs utilize experimental designs such as the Completely Randomized Design (CRD) and Randomized Complete Block Design (RCBD) because these designs help highlight the effects of heterogeneity in field conditions. According to Moore et al. (2023) [12], these designs ensure that differences observed in the experiments are due to genetic effects rather than environmental differences. Werner (2022) [16] emphasizes that well-planned layouts improve the reliability and reproducibility of field experiments. The foundational principles of blocking, randomization, and replication, introduced by Fisher (1935) in *The Design of Experiments* [6], remain central to minimizing bias and ensuring accuracy.

In CRD, treatments are randomly assigned to plots without grouping, making it flexible when there is little to no variation in field conditions [16]. However, the effectiveness of CRD is compromised under high levels of environmental heterogeneity since it does not consider spatial variation or block effects. On the other hand, RCBD accounts for spatial variability by grouping plots into blocks with similar conditions [7]. Modern mixed-model approaches allow further optimization of RCBD by managing complex field gradients and spatial heterogeneity [8], [15]. These studies show that while classical experimental designs in plant breeding remain robust, their effective implementation not only relies on fundamental principles but also requires standardization and reduced human error through computational tools.

B. Genotype × Environment Interactions and Multi-environment Trials

Genotype x Environment (G×E) interactions are of utmost importance in plant breeding, as understanding this will help in identifying stable and high-performing genotypes. As summarized by Pagnotta and Tripodi (2021) [13], the presence of a G×E interaction complicates the interpretation of trial results, therefore careful trial planning and appropriate statistical analysis are required. Several reports including tropical maize and cassava have recently documented the effectiveness of G×E captured using genomic prediction models with factor analytic approaches [1], [5], [11], [17]. These studies identified that the inclusion of multi-environment trials (METs) strengthens selection for adaptability to diversified conditions, hence improving the efficiency of breeding.

C. Digital Tools and Computational Support in Field Trials

Digital tools and platforms in collecting data further standardize data collection process in field trials, reducing human error and enhancing reproducibility in plant breeding experiments. An example of these tools is *Field Book*, an open-source Android application used for recording field data directly with the ability to manage trials effectively [14]. The *agricolae* R package, on the other hand, provides statistical procedures for RCBD, among others, allowing automated layout generation and analysis [4]. Additionaly, online mapping tools, including Leaflet, a library of GIS-based platforms, enable interactive visualizations of field trials, spatial layouts, and environmental heterogeneity [10]. Finally, remote sensing applications support breeders in monitoring variability within the field and making data-driven decisions on plot allocation [2]. Collectively, these tools help enhance precision and efficiency in field trials, yet there are only a small number of integrated systems that combines trial planning, layout generation, field visualization and reporting in a single workflow.

D. Advances in Statistical Methods and Trial Optimization

Modern field trials use advance statistical methods to consider spatial variability and complex environmental factors. Smith and Cullis (2020) [15] discuss further improvements in classic RCBD method through mixed model approaches. These approaches are done in order to account for both fixed effects and random effects in field trials. In addition, spatial designs and multi-environment trials (MET) have been outperforming classical designs through reducing experimental error which leads to better detection of treatment differences [1], [5], [8], [17]. These techniques allow plant breeders to extract precise and meaningful genetic information even in highly variable field conditions.

E. Need for an Integrated Field Layout System

Despite the existence of computational tools and advance digital platforms for plant breeding experiments, current solutions often offer independent features such as layout generation, data collection, or analysis. Additionally, existing methods require significant amount of manual input, which often lead to inconsistencies and errors [4], [14]. In order to account for these issues, an integrated platform that combines automated experimental field layout generation, interactive visualization and standardized report production is needed. With this, improvement in efficiency, reduced human error and reproducibility of trials across different locations and seasons can be achieve.

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F. Synthesis

This literature review shows that while classical experimental designs and advance digital tools provide support for plant breeding trials, there remains a gap in fully integrated system that can handle trial planning, visualization and reporting in one workflow. The development of *LAYA: A Web-Based System for Automated Experimental Field Layout Generation, Visualization, and Planting Plan Report Generation* addresses these limitations. Through automating layout creation, providing an interactive visual map for field trials, and producing standardized reports, LAYA offers a comprehensive solution that aligns with modern plant breeding requirements, ensuring accuracy, reproducibility, and efficiency.

III. Materials and Methods

A. Introduction

This chapter presents the methodologies, technologies, and architectural decisions used in developing LAYA, a web-based system for automated field layout generation, experiment management, map-based visualization, and planting plan report creation. The system integrates user authentication, administrative approval workflows, experiment creation and deletion, experimental design generation (RCBD and CRD), spatial mapping with Leaflet, and trial-level data storage through a PostgreSQL/PostGIS backend.

Figure 4 illustrates the overall system architecture, while Figures 5–10 show the wireframes used in designing the interface.

B. Development and Testing Flow

The development process followed an iterative approach consisting of planning, prototyping, implementation, testing, and feature refinement. Major stages included:

1) **Requirements analysis** with the Feeds and Industrial Crops Section, Institute of Plant Breeding (IPB).
2) **Wireframing** of the user interface, including the map, dashboard, and experiment creation wizard.
3) **System architecture design**, particularly for handling authentication, user roles, experiment management, notifications, and spatial data.
4) **Backend implementation** with Express.js, PostgreSQL, and PostGIS for trial geometry storage.
5) **Frontend development** using React, Leaflet for spatial visualization, and REST API integration.
6) **Testing** using unit tests (backend), UI tests, and scenario testing using sample trials.
7) **Refinements**, including dashboard redesign, notification system, and experiment wizard modification.

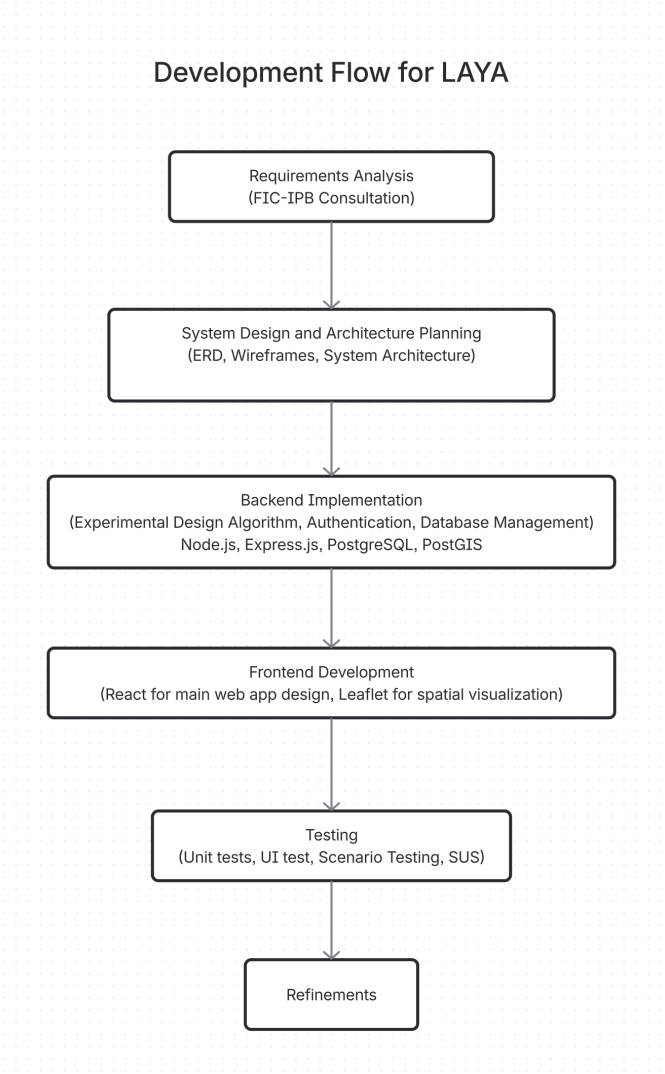

Fig. 3: Development Flow Diagram.

C. Tech Stack and Tools

LAYA was developed using the following tools:

- **Frontend**: React.js, Leaflet.js, TailwindCSS
- **Backend**: Node.js (Express.js), JSON Web Tokens (JWT) for authentication
- **Database**: PostgreSQL with PostGIS extension for spatial storage
- **Mapping**: Leaflet.js + GeoJSON for polygon rendering
- **Algorithms**: Custom RCBD and CRD layout generators
- **PDF Generation**: jsPDF and html2canvas
- **Version Control**: Git and GitHub

These tools were selected for their scalability, open-source availability, and compatibility with modern GIS workflows.

D. System Architecture

LAYA's architecture consists of three major layers: (1) frontend client, (2) backend server, (3) relational-spatial database.

The system supports two user roles: **Admin** and **Staff**. Admin users approve account requests, manage experiments

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globally, and view all trials. Staff members can create experiments, submit deletion requests, and view their own trials.

Key components include:

- **Authentication and Authorization Module** — Handles login, account requests, role-based access, and JWT-secured endpoints.
- **Experiment Management API** — Supports creation, retrieval, deletion (with admin approval), and update of experiments.
- **Design Generation Module** — Produces randomized layouts for RCBD and CRD using user-defined factors such as varieties, replications, plant spacing, rows, and plants per row.
- **Mapping Module (Leaflet + PostGIS)** — Stores, retrieves, and visualizes field polygons representing experiments.
- **Notification Module** — Logs and displays account approvals, deletion status, and experiment updates.
- **Dashboard Module** — Shows experiment list, recent activity logs, and tools to create/delete experiments.

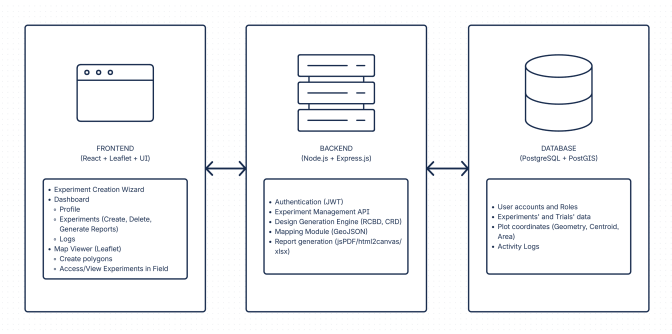

Fig. 4: 3-Layer Architecture of LAYA

E. Features of the Application

1) User Authentication and Access Control: LAYA implements secure login using JWT tokens. Admin approval is required for new users, and every action a user makes is logged for traceability. Notifications are generated for approvals, denials, and experiment updates.

2) Interactive Field Map: After login, users are redirected to the *Map* interface where all experiments are displayed as polygons using Leaflet. Experiments are color-coded and clickable, allowing users to view trial summaries and generate layouts.

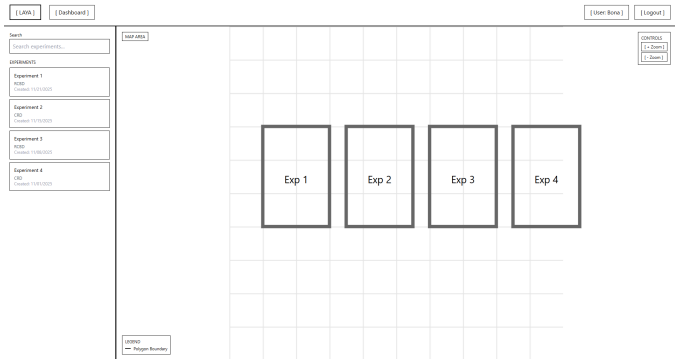

Fig. 5: Wireframe of the LAYA Map Interface.

3) User Dashboard: The Dashboard consolidates user-specific operations:

- Profile details (name, role)
- Notification panel (approval, rejection, experiment status)
- Buttons for *Create Experiment* and *Delete Experiment*
- Table of created experiments with quick access links
- Recent activity logs

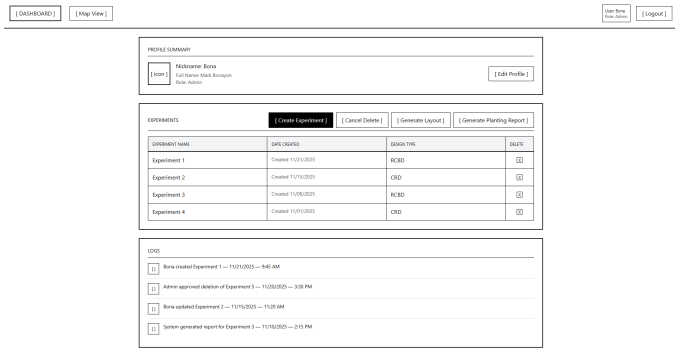

Fig. 6: Wireframe of the user dashboard interface.

4) Experiment Creation Wizard: The experiment creation wizard is designed as a four-step guided workflow that allows users to define experiment parameters, configure trials, review details, and draw the field boundary on the map.

Step 1: Experiment Details

This step collects the general metadata and structural parameters of the experiment:

- Experiment Name
- Design Type (RCBD or CRD)
- Crop
- Number of Trials
- Description
- Replications per Trial
- Varieties per Replication
- Rows per Plot
- Plants per Row
- Plant Spacing
- Row Spacing
- Alleyway Spacing

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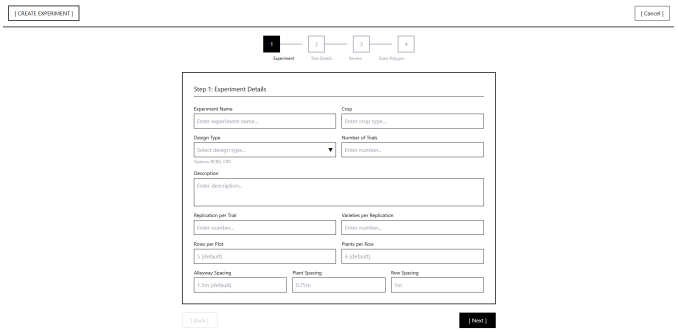

Fig. 7: Wireframe for Step 1: Experiment Details.

Step 2: Trial Details

The user configures each trial individually. Trials displayed in the dropdown are automatically generated based on the number provided in Step 1.

- Trial Selection (dropdown)
- Edit Trial Name
- Enter Variety Names for the selected trial

Fig. 8: Wireframe for Step 2: Trial Details.

Step 3: Review

This step serves as a summary of all inputs before finalizing the experiment. Users can verify the correctness of experiment metadata, trial names, and variety lists before proceeding.

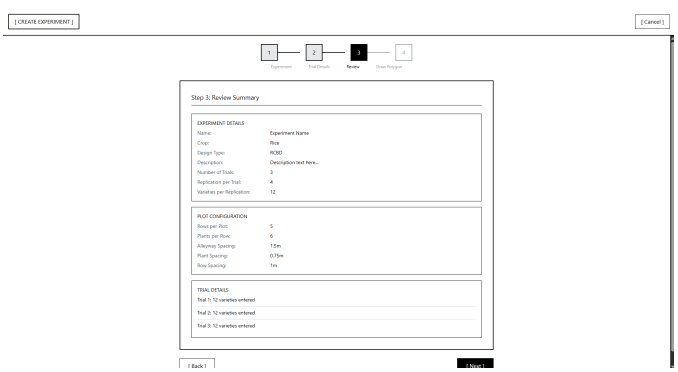

Fig. 9: Wireframe for Step 3: Review.

Step 4: Draw Polygon

The final step prompts the user to draw the experimental field boundary on the map. This polygon defines the physical area where replications and plots will be automatically laid out.

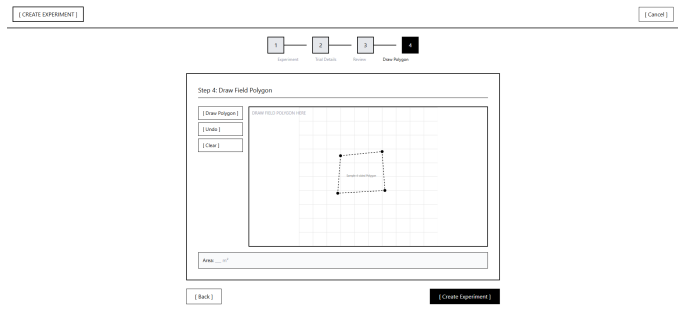

Fig. 10: Wireframe for Step 4: Draw Polygon on the map.

5) Layout Generation and Planting Plan Report: After creating an experiment using the wizard, LAYA allows users to generate and view the pre-randomized layouts and export both the field layout diagrams and planting plan reports. Users select a specific experiment from the Dashboard and use dedicated buttons to produce and export these documents.

- **Per-Experiment Layout Display:** Users select an experiment from the Dashboard and click the *Generate Layout* button. The system displays the pre-randomized plot layouts for the selected experiment, showing replications, plot numbers, and variety assignments in an interactive field diagram. Layouts can be displayed in color or black-and-white depending on user preference.
- **Planting Plan Report:** By clicking the *Generate Planting Report* button, users can export a detailed PDF containing all trials for the selected experiment. The report includes experiment metadata, trial details, plot layout tables, replication information, plant spacing, and any remarks.
- **Export Options:** Both the layout diagrams and planting plan reports can be exported in PDF format for printing or distribution. Future enhancements may include Excel or Google Sheets export for further analysis.

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RCBD Planting Plan

Trial: PYT 2

Location: Tranca-West, IPB

Date Planted: 2025-07-26

Season: Wet Season

4 replications, 16 plots/rep, 5 rows/plot, 7 plants/row

Spacing: 1m x 0.75m | Row Length: 4.25m | Alley: 1.5m

Entry No.	Variety	Rep 1	Rep 2	Rep 3	Rep 4	Remarks
1	CG03-01	3	24	48	59	
2	CG04-02	14	27	42	51	
3	CG05-03	2	29	36	56	
4	CG06-04	10	25	45	62	
5	CG07-05	1	21	44	49	
6	CG08-06	7	20	34	58	
7	CG09-07	12	30	39	60	
8	CG10-08	5	23	47	63	
9	CG11-09	16	19	35	53	
10	CG12-10	6	18	46	55	
11	CG13-11	13	28	43	61	
12	CG14-12	11	31	40	54	
13	CG15-13	15	17	37	50	
14	CG16-14	9	22	33	64	
15	CG17-15	8	32	38	57	
16	CG18-16	4	26	41	52	

Fig. 11: Example of a generated planting plan report showing all trials for a selected experiment.

Fig. 12: Example of a generated layout diagram for a selected experiment. Layout can be color-coded or black-and-white.

F. Wireframes

All wireframes used during development are included for reference and appear in this chapter:

- Map Interface (Figure 5)
- Dashboard Interface (Figure 6)
- Experiment Wizard (Figures 7 8 9 and 10)

G. Testing

The system underwent several stages of testing:

- **Unit Testing**: Layout generation logic (RCBD/CRD), API endpoints, and authentication.
- **Integration Testing**: Frontend–backend communication and database transactions.
- **Usability Testing**: Wizard flow, dashboard interactions, and map navigation.
- **Spatial Validation**: Polygon accuracy and correct trial alignment using Leaflet and PostGIS.
- **Performance Testing**: Rendering multiple experiments on the map and generating large trial layouts.

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